Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The unit circle and radian measure**

Convert 210° to radians.

$210 \times \frac{\pi}{180} = \frac{7\pi}{6}$.

2 **[Foundational]** • **Target: The unit circle and radian measure**

Convert $\frac{5\pi}{4}$ radians to degrees.

$\frac{5\pi}{4} \times \frac{180}{\pi} = 225^\circ$.

3 **[Foundational]** • **Target: The unit circle and radian measure**

Find the exact value of $\sin(240^\circ)$.

240° is in Quadrant III with reference angle 60°, where sine is negative: $\sin(240^\circ) = -\frac{\sqrt{3}}{2}$.

4 **[Foundational]** • **Target: Trigonometric identities**

If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant II, find $\cos\theta$ and $\tan\theta$.

$\cos\theta = -\sqrt{1 - \frac{9}{25}} = -\frac{4}{5}$ (negative in Quadrant II). $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{-4/5} = -\frac{3}{4}$.

5 **[Foundational]** • **Target: Double angle formulas**

Given $\cos\theta = \frac{5}{13}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

$\sin\theta = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}$. $\sin 2\theta = 2\left(\frac{12}{13}\right)\left(\frac{5}{13}\right) = \frac{120}{169}$. $\cos 2\theta = 1 - 2\sin^2\theta = 1 - \frac{288}{169} = -\frac{119}{169}$.

6 **[Foundational]** • **Target: Double angle formulas**

Use double-angle formulas to express $\sin 4\theta$ in terms of $\sin\theta$ and $\cos\theta$ only.

$\sin 4\theta = 2\sin 2\theta\cos 2\theta = 2(2\sin\theta\cos\theta)(1 - 2\sin^2\theta) = 4\sin\theta\cos\theta(1 - 2\sin^2\theta)$.

7 **[Foundational]** • **Target: Graphs of trigonometric functions**

The graph shows $y = 3\sin(2x)$. State the amplitude and period.

Amplitude $= 3$. Period $= \frac{2\pi}{2} = \pi$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

© ClipSAT — clipsat.org | All Rights Reserved

8 [Intermediate] • Target: Graphs of trigonometric functions

The function $y = 4\cos(3x - \pi)$ has what amplitude, period, and phase shift?

Amplitude $= 4$. Period $= \frac{2\pi}{3}$. Writing $3x - \pi = 3\left(x - \frac{\pi}{3}\right)$, the phase shift is $\frac{\pi}{3}$ to the right.

9 [Intermediate] • Target: The Law of Sines

In triangle ABC, $a = 10$, $A = 40^\circ$, $B = 65^\circ$. Find b, to the nearest hundredth.

$\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{10\sin 65^\circ}{\sin 40^\circ} \approx \frac{9.063}{0.643} \approx 14.10$.

10 [Intermediate] • Target: The Law of Sines

In triangle ABC, $a = 8$, $b = 10$, $A = 35^\circ$. Determine how many triangles are possible, and find angle B (to the nearest tenth of a degree) for each.

$h = b\sin A = 10\sin 35^\circ \approx 5.74$. Since $h < a < b$, two triangles are possible.
$\sin B = \frac{b\sin A}{a} \approx 0.717 \Rightarrow B \approx 45.8^\circ$ or $B \approx 180^\circ - 45.8^\circ = 134.2^\circ$. Both are valid since $A + B < 180^\circ$ in each case.

11 [Intermediate] • Target: The Law of Cosines and the sine area formula

In triangle ABC shown below, $a = 11$, $b = 14$, and $C = 55^\circ$. Find side c, to the nearest hundredth, and the area of the triangle, to the nearest tenth.

$c^2 = 11^2 + 14^2 - 2(11)(14)\cos 55^\circ \approx 317 - 176.66 = 140.34$, so $c \approx 11.85$. Area $= \frac{1}{2}(11)(14)\sin 55^\circ = 77\sin 55^\circ \approx 63.1$.

12 [Intermediate] • Target: The Law of Cosines and the sine area formula

State the two formulas used above (Law of Cosines for a side, and the sine formula for area) in general form for a triangle with sides a, b, c and included angle C.

Law of Cosines: $c^2 = a^2 + b^2 - 2ab\cos C$. Area $= \frac{1}{2}ab\sin C$.

13 [Intermediate] • Target: Inverse trigonometric functions

Evaluate $\arcsin\left(\frac{\sqrt{2}}{2}\right)$ exactly, in radians.

$\frac{\pi}{4}$.

14 [Intermediate] • Target: Inverse trigonometric functions

Find $\theta = \arctan(1)$ in degrees (principal value).

45°.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

© ClipSAT — clipsat.org | All Rights Reserved

15 [Challenging] • Target: A well-designed word problem: locating a ship from two lighthouses • Cross-Lesson: 6.01 Circles & Coordinate Geometry

This question has three parts. Two lighthouses A and B are 15 km apart along a straight coastline. A ship at point C is sighted such that $\angle CAB = 72^\circ$ and $\angle CBA = 58^\circ$ (both measured from line AB toward the ship). (a) Find $\angle ACB$. (b) Use the Law of Sines to find the distance AC from lighthouse A to the ship, to the nearest tenth of a km. (c) Find the distance BC from lighthouse B to the ship, to the nearest tenth of a km.

(a) $\angle ACB = 180^\circ - 72^\circ - 58^\circ = 50^\circ$. (b) AC is opposite $\angle B$: $AC = \frac{AB \sin B}{\sin C} = \frac{15 \sin 58^\circ}{\sin 50^\circ} \approx 16.6$ km. (c) BC is opposite $\angle A$: $BC = \frac{AB \sin A}{\sin C} = \frac{15 \sin 72^\circ}{\sin 50^\circ} \approx 18.6$ km.

16 [Challenging] • Target: The Law of Cosines and the sine area formula • Cross-Lesson: 7.01 Solid Geometry & Similarity

In triangle ABC, $a = 8$, $\angle B = 35^\circ$, and $\angle C = 75^\circ$. Use the Law of Sines to calculate the length of side b to two decimal places.

Step 1: Find angle A: $\angle A = 180^\circ - 35^\circ - 75^\circ = 70^\circ$. Step 2: Apply Law of Sines: $\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{a \sin B}{\sin A} = \frac{8 \sin(35^\circ)}{\sin(70^\circ)}$. Step 3: Evaluate: $b = \frac{8(0.5736)}{0.9397} = \frac{4.5886}{0.9397} \approx \mathbf{4.88}$.

17 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 8.01 Statistics & Probability

Two sides of a right triangle are 7 cm and 10 cm. If the third side is the hypotenuse, find: (a) the exact hypotenuse length, and (b) the measure of the smaller acute angle to the nearest tenth of a degree.

(a) Exact hypotenuse: $c = \sqrt{7^2 + 10^2} = \sqrt{49 + 100} = \sqrt{\mathbf{149}}$ cm ≈ 12.21 cm. (b) Smaller acute angle: The smaller angle is opposite the shorter leg (7 cm): $\tan\theta = \frac{7}{10} = 0.7 \Rightarrow \theta = \arctan(0.7) \approx \mathbf{35.0}^\circ$.

18 [Challenging] • Target: A well-designed word problem: locating a ship from two lighthouses • Cross-Lesson: 1.01 Advanced Algebra

[Unit Circle & Trigonometric Identities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

19 [Challenging] • Target: The unit circle and radian measure • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.

Step 1: Identify smallest angle: The smallest angle is A, opposite the shortest side $a = 7$. Step 2: Law of Cosines for angle A: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{9^2 + 10^2 - 7^2}{2(9)(10)} = \frac{81 + 100 - 49}{180} = \frac{132}{180} = \frac{11}{15} \approx 0.7333$. $A = \arccos(11/15) \approx 42.8^\circ$. Step 3: Heron's formula for area: Semi-perimeter $s = \frac{7 + 9 + 10}{2} = 13$..

© ClipSAT — clipsat.org | All Rights Reserved

20 [Challenging] • Target: Trigonometric identities • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A regular hexagon has side length $s = 6$ cm. (a) Calculate the exact area of the hexagon. (b) Find the length of the long diagonal connecting two opposite vertices.

(a) Exact area: A regular hexagon is composed of 6 congruent equilateral triangles of side length $s = 6$. The area of one equilateral triangle is $\frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}(36) = 9\sqrt{3}$. Total area is: $A = 6 \times 9\sqrt{3} = \mathbf{54\sqrt{3}}$ cm^2 ≈ 93.53 cm^2. (b) Long diagonal: The long diagonal passes through the center of the circumscribed circle and equals twice the side length: $d = 2s = 2(6) = \mathbf{12}$ cm.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The unit circle and radian measure

In triangle ABC, $a = 9$, $b = 15$, and $C = 110^\circ$. Find side c, to the nearest hundredth, and the area, to the nearest tenth.

a. $18.96, 61.4$

b. $19.96, 65.4$

c. $20.96, 63.4$

d. $19.96, 63.4$ ✓

$c^2 = 9^2 + 15^2 - 2(9)(15)\cos 110^\circ \approx 306 + 92.35 \approx 398.35$, so $c \approx 19.96$. Area $= \frac{1}{2}(9)(15)\sin 110^\circ = \frac{135}{2}\sin 110^\circ \approx 63.4$.

22 [Foundational] • Target: Trigonometric identities

In ΔABC, $\angle A = 30^\circ$, $\angle B = 90^\circ$, and side $a = 6$. Find side b (Law of Sines).

a. **12** ✓

b. 13

c. 2

d. 3

$\frac{a}{\sin A} = \frac{b}{\sin B} \Rightarrow b = a \cdot \frac{\sin 90^\circ}{\sin 30^\circ} = 6 \cdot \frac{1}{1/2} = 12$.

23 [Foundational] • Target: Double angle formulas

In ΔABC, $\angle A = 90^\circ$, $a = 10$, and $b = 5$, with $\angle B$ acute. Find $\angle B$ (Law of Sines).

a. 118

b. 89

c. 53

d. **30** ✓

$\frac{\sin B}{b} = \frac{\sin A}{a} \Rightarrow \sin B = \frac{b \sin A}{a} = \frac{5 \times 1}{10} = 1/2 \Rightarrow \angle B = 30^\circ$ (acute).

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Solving trigonometric equations

In ΔABC, $a = 12$, $b = 9$, $c = 15$. Find $\angle C$ (Law of Cosines).

a. 47

b. 17

c. **90** ✓

d. 18

$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{144 + 81 - 225}{2(12)(9)} = 0 \Rightarrow \angle C = 90^\circ$.

25 [Foundational] • Target: Graphs of trigonometric functions

In ΔABC, $a = 8$, $b = 5$, and $\angle C = 60^\circ$. Find side c (Law of Cosines).

a. 1

b. 19

c. **7** ✓

d. 3

$c^2 = a^2 + b^2 - 2ab\cos C = 8^2 + 5^2 - 2(8)(5)(\frac{1}{2}) = 49 \Rightarrow c = 7$.

26 [Foundational] • Target: The Law of Sines

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$ ✓

c. 1

d. $\frac{\sqrt{2}}{2}$

The equation of a unit circle centered at the origin is $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $y = \frac{\sqrt{3}}{2}$. Substituting this into the equation of the unit circle: $x^2 + (\frac{\sqrt{3}}{2})^2 = 1$ $x^2 + \frac{3}{4} = 1$ $x^2 = 1 - \frac{3}{4}$ $x^2 = \frac{1}{4}$ Since point P lies in the first quadrant, its x-coordinate must be positive. Therefore, $x = \sqrt{\frac{1}{4}} = \frac{1}{2}$. Alternatively, we are given that the angle θ formed by the ray from the or.

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: The Law of Cosines and the sine area formula

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?

a. $\cos\theta$ ✓

b. $\sin\theta$

c. $\tan\theta$

d. 1

By definition, the coordinates of a point $P(x, y)$ on the unit circle corresponding to an angle θ (measured counterclockwise from the positive x-axis) are given by $(\cos\theta, \sin\theta)$. Therefore, the x-coordinate of the point P is $\cos\theta$.

28 [Foundational] • Target: Inverse trigonometric functions

Using a double-angle identity, evaluate $\cos^2\theta - \sin^2\theta$ when $\theta = 15^\circ$.

a. -1

b. $-\frac{\sqrt{3}}{2}$

c. $\frac{\sqrt{2}}{2}$

d. $\frac{\sqrt{3}}{2}$ ✓

$\cos^2\theta - \sin^2\theta = \cos(2\theta)$; at $\theta = 15^\circ$, $2\theta = 30^\circ$, so the value is $\frac{\sqrt{3}}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: locating a ship from two lighthouses

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$ ✓

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

We are given $\cos\alpha = \frac{3}{5}$ and α is in the first quadrant. Using the identity $\sin^2\alpha + \cos^2\alpha = 1$, we find $\sin\alpha = \sqrt{1 - (\frac{3}{5})^2} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$ (since α is in the first quadrant, $\sin\alpha$ is positive). We are given $\sin\beta = -\frac{4}{5}$ and β is in the third quadrant. Using the identity $\sin^2\beta + \cos^2\beta = 1$, we find.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: The unit circle and radian measure

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

The problem states that point $P(x, y)$ lies on the unit circle, meaning $x^2 + y^2 = 1$. We are given that the y-coordinate of P is $y = \frac{3}{5}$. Substituting this into the unit circle equation, we get $x^2 + (\frac{3}{5})^2 = 1$. This simplifies to $x^2 + \frac{9}{25} = 1$. Subtracting $\frac{9}{25}$ from both sides gives $x^2 = 1 - \frac{9}{25} = \frac{16}{25}$. Taking the square root of both sides, we get $x = \pm\frac{4}{5}$. The problem also states that point P is in the second quadrant, where the x.

31 [Intermediate] • Target: Trigonometric identities

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$ ✓

d. $\frac{4}{5}$

On the unit circle, any point P(x, y) satisfies the equation $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $\frac{3}{5}$ and P is in the second quadrant. For a point $P(\cos\theta, \sin\theta)$ on the unit circle, we have $x = \cos\theta$ and $y = \sin\theta$. Thus, we have $\sin\theta = \frac{3}{5}$. Substituting this into the unit circle equation: $\cos^2\theta + (\frac{3}{5})^2 = 1$. $\cos^2\theta + \frac{9}{25} = 1$. $\cos^2\theta = 1 - \frac{9}{25}$. $\cos^2\theta = \frac{16}{25}$..

32 [Intermediate] • Target: Double angle formulas

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$ ✓

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

For any point on the unit circle, the coordinates are $(\cos(\theta), \sin(\theta))$. The equation of the unit circle is $x^2 + y^2 = 1$. Substituting the coordinates, we get $\cos^2(\theta) + \sin^2(\theta) = 1$. We are given that $\cos(\theta) = \frac{1}{2}$. So, $\left(\frac{1}{2}\right)^2 + \sin^2(\theta) = 1$. This simplifies to $\frac{1}{4} + \sin^2(\theta) = 1$. Subtracting $\frac{1}{4}$ from both sides gives $\sin^2(\theta) = 1 - \frac{1}{4} = \frac{3}{4}$. Taking the square root of both sides, we get.

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Solving trigonometric equations

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. **1** ✓

d. 5/3

The point P is on the unit circle, so its coordinates are $(\cos\theta, \sin\theta)$. We are given $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. The point Q has coordinates (x, y) and its distance from the origin is 5. The angle between the positive x-axis and OQ is α. Thus, $x = 5\cos\alpha$ and $y = 5\sin\alpha$. We are asked to find the maximum possible value of the product of the x-coordinates of P and Q, which is $\cos\theta \cdot x$. Substituting the given values, we h.

34 [Intermediate] • Target: Graphs of trigonometric functions

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. Given that AC = 5, BC = 12, and angle BAC = θ, find the value of $\sin(\theta)$.

a. $\frac{12}{13}$ ✓

b. $\frac{5}{13}$

c. $\frac{12}{5}$

d. $\frac{5}{12}$

In the right-angled triangle ABC, the side opposite to angle θ (angle BAC) is BC, and the hypotenuse is AB. We are given AC = 5 and BC = 12. We can find the length of the hypotenuse AB using the Pythagorean theorem: $AB^2 = AC^2 + BC^2$. $AB^2 = 5^2 + 12^2 = 25 + 144 = 169$. Therefore, $AB = \sqrt{169} = 13$. The sine of an angle in a right-angled triangle is defined as the ratio of the length of the opposite side to the length of the hypotenuse. So.

35 [Intermediate] • Target: The Law of Sines

In the right-angled triangle ABC shown below, the length of the side adjacent to angle A is 12 units and the length of the hypotenuse is 13 units. What is the cosine of angle A?

a. $\frac{5}{13}$

b. $\frac{12}{13}$ ✓

c. $\frac{5}{12}$

d. $\frac{13}{5}$

In a right-angled triangle, the cosine of an angle is defined as the ratio of the length of the adjacent side to the length of the hypotenuse. For angle A in triangle ABC, the adjacent side is given as 12 units and the hypotenuse is given as 13 units. Therefore, $\cos(A) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{12}{13}$.

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: The Law of Cosines and the sine area formula

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$ ✓

d. 1

The coordinates of a point on the unit circle corresponding to an angle θ are given by $(\cos\theta, \sin\theta)$. In this case, $\theta = \frac{\pi}{3}$. Therefore, the y-coordinate of point P is $\sin\left(\frac{\pi}{3}\right)$. We know that $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$.

37 [Intermediate] • Target: Inverse trigonometric functions

State the double-angle formula $\sin 2\theta =$

a. $2\cos^2\theta - 1$

b. $\sin^2\theta - \cos^2\theta$

c. $2\sin\theta\cos\theta$ ✓

d. $\cos^2\theta - \sin^2\theta$

Applying Trigonometric Functions principles, Double angle formula. Therefore, the verified correct option is $2\sin\theta\cos\theta$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: locating a ship from two lighthouses • Cross-Lesson: 1.01 Advanced Algebra

In the diagram, ΔABC is a right-angled triangle with the right angle at C. The length of the hypotenuse AB is 26 units. Point D lies on AB such that CD is the altitude to the hypotenuse. If $AD = 4$ units and $BD = x$ units, find the value of x.

a. 12

b. 13

c. 16 ✓

d. 22

In a right-angled triangle, the altitude to the hypotenuse creates similar triangles. Specifically, $\Delta ADC \sim \Delta CDB \sim \Delta ACB$. Using the property that the square of the altitude is equal to the product of the segments of the hypotenuse (geometric mean theorem), we have $CD^2 = AD \times BD$. Alternatively, we can use the similarity between ΔADC and ΔCDB: $\frac{AD}{CD} = \frac{CD}{BD}$ which leads to $CD^2 = AD \times BD$. We also know that in a right-angled triangle, the squ.

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: The unit circle and radian measure • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

In the diagram, two chords AB and CD intersect at point P inside a circle. If $PA = 6$, $PB = 8$, and $PC = 4$, what is the length of PD?

a. 6

b. 8

c. **12** ✓

d. 10

By the Power of a Point theorem (Intersecting Chords Theorem), for two chords intersecting inside a circle, the product of the segments of one chord is equal to the product of the segments of the other chord. Therefore, we have $PA \cdot PB = PC \cdot PD$. Substituting the given values: $6 \cdot 8 = 4 \cdot PD$. This simplifies to $48 = 4 \cdot PD$. Dividing both sides by 4, we get $PD = \frac{48}{4} = 12$.

40 [Challenging] • Target: Trigonometric identities • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Consider a circle with center O. Two chords AB and CD intersect at point P inside the circle. If AP = 3, PB = 8, and CP = 4, determine the length of PD. The diagram shows a circle with intersecting chords.

a. **6** ✓

b. 8

c. 10

d. 12

According to the Power of a Point theorem for intersecting chords, the product of the segments of one chord is equal to the product of the segments of the other chord. Therefore, we have AP · PB = CP · PD. Substituting the given values: 3 · 8 = 4 · PD 24 = 4 · PD PD = $\frac{24}{4}$ PD = 6 Thus, the length of PD is 6.

41 [Challenging] • Target: Double angle formulas • Cross-Lesson: 4.01 Lines, Triangles & Polygons

In ΔABC, $\angle A = 60^\circ$, $\angle B = 80^\circ$, and side $a = 5$. Find side b (to one decimal place) using the Law of Sines.

a. **5.7** ✓

b. 6.7

c. 7.7

d. 4.7

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{5 \sin 80^\circ}{\sin 60^\circ} \approx 5.7$.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Solving trigonometric equations • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Two similar solids have a surface-area ratio of $25:9$. Find their volume ratio.

a. 125:1

b. **125:27** ✓

c. 125:729

d. 125:216

Linear ratio $=\sqrt{25:9}=5:3$. Volume ratio $=5^3:3^3=125:27$.

43 [Challenging] • Target: Graphs of trigonometric functions • Cross-Lesson: 7.01 Solid Geometry & Similarity

In the diagram below, a circle has its center at the origin $(0,0)$ and passes through the point $(3,4)$. What is the radius of the circle?

a. **5** ✓

b. 7

c. 25

d. 12

The equation of a circle centered at the origin is $x^2+y^2=r^2$, where r is the radius. Since the circle passes through the point $(3,4)$, we can substitute these coordinates into the equation: $3^2+4^2=r^2$. This simplifies to $9+16=r^2$, so $r^2=25$. Taking the square root of both sides, we get $r=5$ (since the radius must be positive). Therefore, the radius of the circle is 5.

44 [Challenging] • Target: The Law of Sines • Cross-Lesson: 8.01 Statistics & Probability

In a right-angled triangle, one of the acute angles is 30°. What is the measure of the other acute angle?

a. 30°

b. 45°

c. 60° ✓

d. 90°

In a right-angled triangle, the sum of the three angles is 180°. One angle is 90° (the right angle), and another acute angle is given as 30°. Let the other acute angle be x. Therefore, we have the equation: $90^\circ+30^\circ+x=180^\circ$. Subtracting 120° from both sides gives $x=180^\circ-120^\circ=60^\circ$.

© ClipSAT — clipsat.org | All Rights Reserved

45 **[Challenging]** • **Target: The Law of Cosines and the sine area formula** • **Cross-Lesson: 1.01 Advanced Algebra**

Prove the identity $\sin 3A = 3\sin A - 4\sin^3 A$ using the angle addition formula.

a. Not an identity

b. Only for $A < \pi/2$

c. Use calculus

d. $\sin(2A + A) = \sin 2A\cos A + \cos 2A\sin A = 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A = 3\sin A - 4\sin^3 A$. ✓ ✓

Applying Trigonometric Functions principles, Standard derivation. Therefore, the verified correct option is $\sin(2A + A) = \sin 2A\cos A + \cos 2A\sin A = 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A = 3\sin A - 4\sin^3 A$. ✓.

© ClipSAT — clipsat.org | All Rights Reserved